

The Torque-free Axisymmetric Rigid body

Most spacecraft designed to be spun are axisymmetric, that is, two of the principal moments of inertia are the same. Also, a satellite in orbits is an almost perfectly torque-free body, considerably simplifying the analysis. Then, Euler equation become ,

$$\begin{aligned} M_1 &= A\dot{\omega}_1 + (C - B)\omega_2\omega_3 \\ M_2 &= B\dot{\omega}_2 + (A - C)\omega_1\omega_3 \\ M_3 &= C\dot{\omega}_3 + (B - A)\omega_1\omega_2 \end{aligned} \Rightarrow \begin{cases} M_1 = 0 = A\dot{\omega}_1 + (C - A)\omega_2\omega_3 \\ M_2 = 0 = A\dot{\omega}_2 + (A - C)\omega_1\omega_3 \\ M_3 = 0 = C\dot{\omega}_3 + (A - A)\omega_1\omega_2 \end{cases} \quad (1)$$

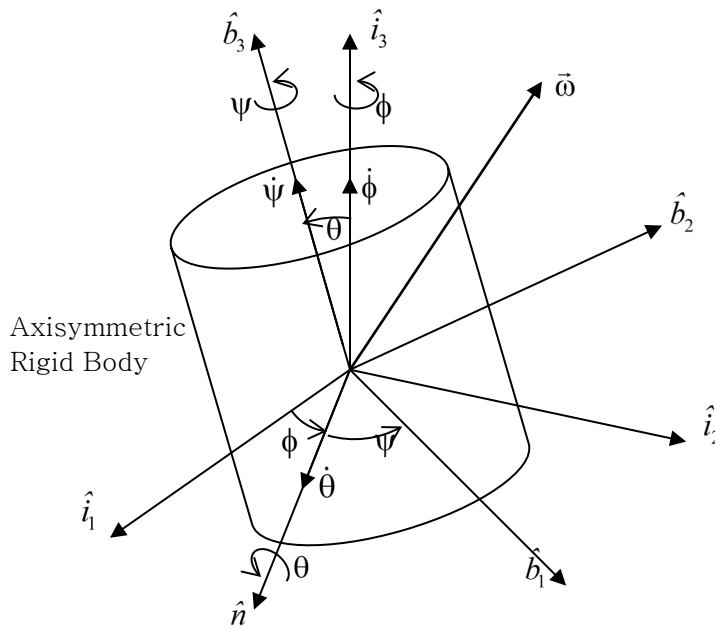
$C = I_3$ is the principal moment of inertia about the longitudinal $\hat{b}_3$ -axis at the cylinder.

$A = I_1 = I_2$ is the centroidal principal moment of inertia about the transverse axis of the cylinder.

$$I = \begin{bmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C \end{bmatrix}$$

(i_1, i_2, i_3) inertial frame, (b_1, b_2, b_3) principal-body frame

$R^{ib} = R_3(\psi)R_1(\theta)R_3(\phi)$: “3-1-3 convention”



The third Euler's equation above becomes,

$$\dot{\omega}_3 = 0 \quad \text{or,} \quad \omega_3 = \Omega \text{ (constant)} \quad (2)$$

The first and second Euler's equations are

$$\dot{\omega}_1 = \frac{A-C}{A} \omega_2 \omega_3 = \frac{A-C}{A} \omega_2 \Omega \quad (3)$$

$$\dot{\omega}_2 = -\frac{A-C}{A} \omega_1 \omega_3 = -\frac{A-C}{A} \omega_1 \Omega \quad (4)$$

These are a pair of constant-coefficient linear differential equations. Differentiates eq. (3) and substitute for $\dot{\omega}_2$ from eq. (4).

$$\begin{aligned} \ddot{\omega}_1 &= \frac{A-C}{A} \Omega \dot{\omega}_2 = \left(\frac{A-C}{A} \right) \left(-\frac{A-C}{A} \right) \Omega^2 \omega_1 \\ \rightarrow \ddot{\omega}_1 + \alpha^2 \omega_1 &= 0 \quad \text{where} \quad \alpha \equiv \frac{C-A}{A} \Omega \end{aligned} \quad (5)$$

The solution of this equation is of the form

$$\omega(t) = a \cos \alpha t + b \sin \alpha t \quad (6)$$

where a and b are some appropriate constants determined from initial conditions

$$\text{or} \quad \omega_1 = \omega_o \cos[\alpha(t-t_o)]$$

where ω_o is the amplitude of the oscillation and the constant t_o allow an arbitrary initial phase

Similarly, differentiates eq. (4) and substitute for $\dot{\omega}_2$ from eq. (3).

$$\begin{aligned} \ddot{\omega}_2 &= -\left(\frac{A-C}{A} \right) \Omega \dot{\omega}_1 = -\left(\frac{A-C}{A} \right) \left(\frac{A-C}{A} \right) \Omega^2 \omega_2 \\ \ddot{\omega}_2 + \alpha \omega_2 &= 0 \quad \text{where} \quad \alpha \equiv \frac{C-A}{A} \Omega \\ \rightarrow \omega_2(t) &= c \cos \alpha t + d \sin \alpha t \\ \text{or} \quad \omega_2(t) &= \omega_o \sin[\alpha(t-t_o)] \end{aligned} \quad (7)$$

Hence, the angular velocity vector obtained from the Euler's equation becomes

$$\begin{cases} \omega_1(t) = \omega_o \cos[\alpha(t - t_o)] \\ \omega_2(t) = \omega_o \sin[\alpha(t - t_o)] \\ \omega_3(t) = \Omega \end{cases} \quad \text{represented in the body frame}$$

$$Eq(3) \times \omega_1 + Eq(4) \times \omega_2$$

$$\begin{aligned} \rightarrow & \left[\dot{\omega}_1 \omega_1 = \left(\frac{A-C}{A} \right) \Omega \omega_1 \omega_2 \right] \\ & + \left[\dot{\omega}_2 \omega_2 = - \left(\frac{A-C}{A} \right) \Omega \omega_1 \omega_2 \right] \end{aligned}$$

$$\text{Thus, } \dot{\omega}_2 \omega_2 + \dot{\omega}_1 \omega_1 = 0$$

$$\frac{d}{dt} [\omega_1^2 + \omega_2^2] = 0$$

$$\therefore \omega_1^2 + \omega_2^2 = \omega_o^2 = \text{constant} \quad \text{where } \omega_o = \sqrt{\omega_1^2 + \omega_2^2}$$

$$\text{Hence, } \omega = \sqrt{\omega_1^2 + \omega_2^2 + \omega_3^2} = \sqrt{\omega_o^2 + \Omega^2} = \text{constant}$$

$\Rightarrow$ The $\vec{\omega}$ vector has a constant projection on satellite's symmetry axis, while the projection of the $\vec{\omega}$ vector on the satellite's plane of symmetry executes a circular motion, with angular frequency, α .

The $\vec{\omega}$ vector thus describes a cone about the satellite's axis of symmetry, which explains the term coning motion applied to the solution.

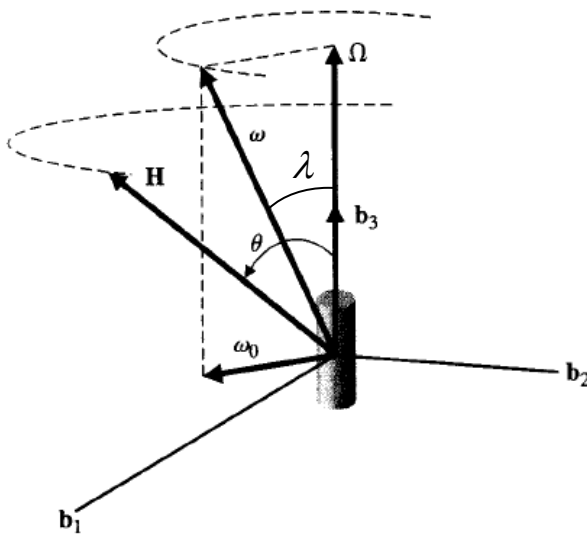


FIGURE 5.1
Angular velocity and angular momentum in the body frame.

$$\hat{b}_3 \cdot \bar{\omega} = 1|\bar{\omega}| \cos \lambda \rightarrow \cos \lambda = \frac{\hat{b}_3 \cdot \bar{\omega}}{|\bar{\omega}|} = \frac{\Omega}{\sqrt{\omega_o^2 + \Omega^2}} = \text{constant} \rightarrow \lambda = \text{constant}$$

Since $\vec{H} = I\bar{\omega}$,

$$\begin{aligned} \vec{H} &= \begin{pmatrix} A & 0 & 0 \\ 0 & A & 0 \\ 0 & 0 & C \end{pmatrix} \begin{pmatrix} \omega_o \cos[\alpha(t-t_o)] \\ \omega_o \sin[\alpha(t-t_o)] \\ \Omega \end{pmatrix} \\ &= \begin{pmatrix} A\omega_o \cos[\alpha(t-t_o)] \\ A\omega_o \sin[\alpha(t-t_o)] \\ C\Omega \end{pmatrix} \quad \text{expressed in body-frame} \end{aligned}$$

We are ultimately in a description of the motion of the satellite seen from the inertial frame.

The satellite has no applied torque ($\vec{M} = 0 \rightarrow \dot{\vec{H}} = 0$), so the angular momentum vector is constant with respect to the inertial frame. This is not true when viewed from the body-frame.

Since the components of the $\bar{\omega}$ vector have simply been multiplied by factors of A, A , and C to produce $\vec{H}$, the $\vec{H}$ vector also cones about the satellite symmetric axis, again with angular frequency α . The angular-velocity vector $\bar{\omega}$ and the angular momentum vector $\vec{H}$ do not coincide.

The angle between the $\vec{H}$ and $\hat{b}_3$ is given by

$$\begin{aligned} \vec{H} \cdot \hat{b}_3 &= 1|\vec{H}| \cos \theta \\ \rightarrow \cos \theta &= \frac{\vec{H} \cdot \hat{b}_3}{|\vec{H}|} = \frac{C\Omega}{\sqrt{A^2\omega_o^2 + C^2\Omega^2}} = \text{constant} \end{aligned}$$

So, θ is constant and depends on ω_o and Ω .

Introduce the Euler angles, ϕ, θ, ψ and align the angular momentum vector $\vec{H}$ with inertial axis $\hat{i}_3$. $\vec{H}$ is invariant being viewed from inertial frame on the torque-free circumstance.

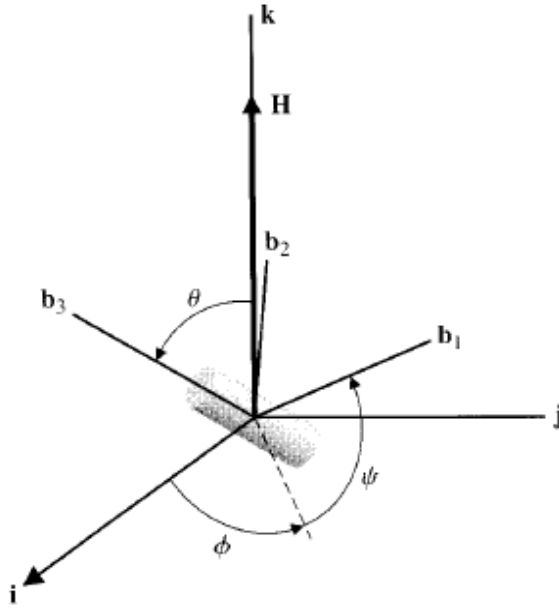


FIGURE 5.2
Satellite motion with respect to the inertial frame.

The Euler angle θ is constant. $\rightarrow \dot{\theta} = 0$

The classical Euler angle rates $(\dot{\phi}, \dot{\theta}, \dot{\psi})$ are related with the angular velocity by

$$\omega_1 = \dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi = \dot{\phi} \sin \theta \sin \psi = \omega_o \cos \alpha t \quad (8)$$

$$\omega_2 = \dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi = \dot{\phi} \sin \theta \cos \psi = \omega_o \sin \alpha t \quad (9)$$

$$\omega_3 = \dot{\psi} + \dot{\phi} \cos \theta = \Omega \quad (10)$$

$$\rightarrow \omega_1^2 + \omega_2^2 = \dot{\phi}^2 \sin^2 \theta = \omega_o^2 \Rightarrow \dot{\phi} = \text{constant} \quad \text{“the precession rate”}$$

$$\dot{\omega}_1 = \ddot{\phi} \sin \theta \sin \psi + \dot{\phi} \cos \theta \dot{\theta} \sin \psi + \dot{\phi} \sin \theta \cos \psi \dot{\psi} = \dot{\phi} \sin \theta \cos \psi \dot{\psi}$$

$$\dot{\omega}_2 = \ddot{\phi} \sin \theta \cos \psi + \dot{\phi} \cos \theta \dot{\theta} \cos \psi - \dot{\phi} \sin \theta \sin \psi \dot{\psi} = -\dot{\phi} \sin \theta \sin \psi \dot{\psi}$$

$$\dot{\omega}_3 = \ddot{\psi} + \ddot{\phi} \cos \theta - \dot{\phi} \sin \theta \dot{\theta} = 0$$

Substituting these equations into the first equation of eq. (1), we obtain

$$0 = A\dot{\omega}_1 + (C - A)\omega_2\omega_3$$

$$0 = A\dot{\phi} \sin \theta \cos \psi \dot{\psi} + (C - A)\dot{\phi} \sin \theta \cos \psi (\dot{\psi} + \dot{\phi} \cos \theta)$$

$$0 = A\dot{\psi} + (C - A)(\dot{\psi} + \dot{\phi} \cos \theta)$$

$$0 = A\dot{\psi} + (C - A)\dot{\phi} \cos \theta + C\dot{\psi} - A\dot{\psi}$$

$$(C - A)\dot{\phi} \cos \theta + C\dot{\psi} = 0$$

$$\dot{\phi} = \frac{C\dot{\psi}}{(A - C) \cos \theta}$$

$$\therefore \dot{\phi} = \frac{C\dot{\psi}}{(A - C) \cos \theta} = \text{constant} \quad (12)$$

$$(\leftarrow \dot{\psi} = \text{const.}, \theta = \text{const.})$$

This expresses the precession rate $\dot{\phi}$ in terms of the inertial spin rate $\dot{\psi}$. $\dot{\phi}$ is the magnitude of the torque-free precession of the $\hat{b}_3$ -axis about $\vec{H}$

→ (i) If $A > C$,

The satellite is a prolate rigid-body, like a slim cylinder, and the precession and spin will occur in the same direction.

→ precession is prograde.

→ (ii) If $A < C$,

The satellite is an oblate rigid-body, like a squat cylinder, and the precession is in the opposite sense from the spin.

→ precession is retrograde.

Since the center of mass is fixed, the kinetic energy is all rotational and constant.

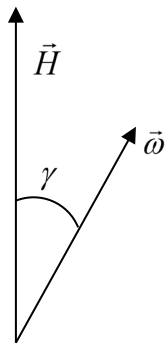
$$T_{rot} = \frac{1}{2} \vec{\omega} \cdot \vec{H} = \frac{1}{2} \{ \vec{\omega}^T \} \{ I \} \{ \vec{\omega} \} = \text{const.}$$

We know that $\vec{H}$ is constant in the inertial-frame

⇒ T_{rot} will be constant only if $\vec{\omega}$ moves in such a way that its projection on the angular momentum vector $\vec{H}$ is constant.

$$\vec{\omega} \cdot \vec{H} = \omega H \cos \gamma$$

$$\rightarrow \cos \gamma = \frac{\vec{\omega} \cdot \vec{H}}{\omega H} = \frac{2T_{rot}}{\omega H} = \text{const.} \quad (\leftarrow T_{rot}, \omega, H \text{ are constant})$$



$\Rightarrow$ Angle γ remains constant and is the half-angle of space-cone.

$\Rightarrow$ The space-cone is the result of precession of $\vec{\omega}$ about the constant angular momentum $\vec{H}$ as viewed from the inertial frame.

① $\vec{H}$, $\vec{\omega}$, and $\hat{b}_3$ - axis all lie in one plane

$$\rightarrow \vec{H} \cdot (\vec{\omega} \times \hat{b}_3) = \det \begin{pmatrix} \vec{H} \\ \vec{\omega} \\ \hat{b}_3 \end{pmatrix} = \begin{vmatrix} A\omega_o \cos[\alpha(t-t_o)] & A\omega_o \sin[\alpha(t-t_o)] & C\Omega \\ \omega_o \cos[\alpha(t-t_o)] & \omega_o \sin[\alpha(t-t_o)] & \Omega \\ 0 & 0 & 1 \end{vmatrix} = 0$$

② $\vec{\omega}$ precesses around $\vec{H}$ with angle γ when viewed in the inertial frame.

$\rightarrow$ This trajectory can be represented as the space-cone.

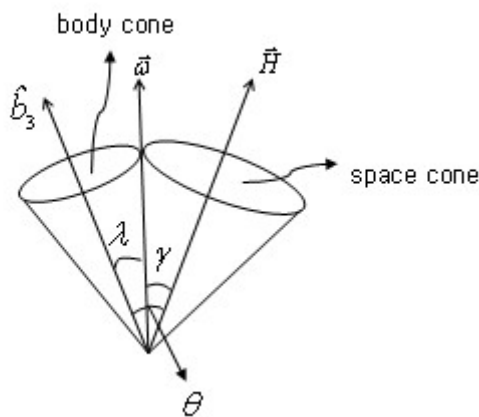
③ $\vec{\omega}$ also precesses around $\hat{b}_3$ (symmetry axis) with angle λ when viewed in the body-frame $\rightarrow$ This trajectory can be represented as the body-cone.

The situation is described as one cone rolling on another. The body cone is rolling without slipping around the space cone and the line of contact is the direction of the $\vec{\omega}$.

i) $A > C$; direct precession

The satellite is prolate, and the precession and spin will occur in the same direction.

The space cone is outside the body cone for the prolate case.



The body cone rolls on the outside surface of space cone.

ii) $A < C$; retrograde precession

The satellite is oblate, and the precession and spin will occur in the opposite direction.

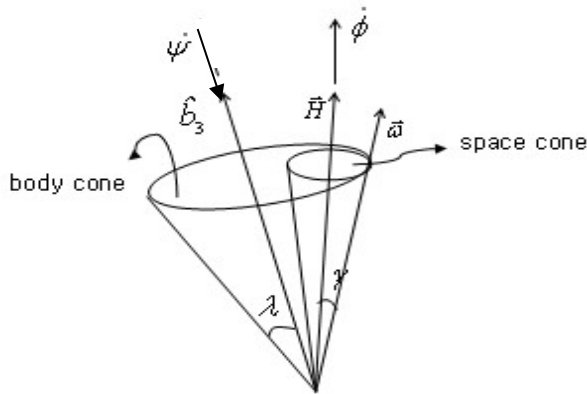
There exist two different cases in the oblate satellite case.

(a) $\theta + \gamma = \lambda$, $\lambda > \gamma$

The space cone is in the inside surface of the body cone which rolls on the outside surface of space cone.

(b) $\gamma = \theta + \lambda$, $\gamma > \lambda$

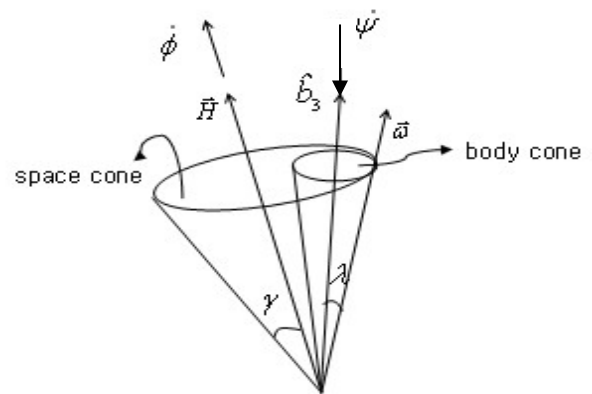
The body cone is inside the space cone.



$$\lambda > \gamma$$

$$\lambda = \theta + \gamma$$

(a)



$$\gamma > \lambda$$

$$\gamma = \theta + \lambda$$

(b)

The body cone rolls on the fixed space cone. In this sense, space cone is in the inside surface of the cone which rolls on the outside(or inside) surface of space cone. $\vec{\omega}$ moves in the opposite sense relative to the body, as compared to its motion case i).